

An Analysis of Convolutional neural network

Convolutional neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

Convolutional neural network -- Part 1

Dropout Because networks have so many parameters, they are prone to overfitting. One method to reduce overfitting is dropout, introduced in 2014. At each training stage, individual nodes are either "dropped out" of the net (ignored) with probability $1 - p$ or kept with probability p , so that a reduced network is left; incoming and outgoing edges to a dropped-out node are also removed. Only the reduced network is trained on the data in that stage. The removed nodes are then reinserted into the network with their original weights. In the training stages, p is usually 0.5; for input nodes, it is typically much higher because information is directly lost when input nodes are ignored. At testing time after training has finished, we would ideally like to find a sample average of all possible 2^n dropped-out networks; unfortunately this is unfeasible for large values of n . However, we can find an approximation by using the full network with each node's output weighted by a factor of p , so the expected value of the output of any node is the same as in the training stages. This is the biggest contribution of the dropout method: although it effectively generates 2^n neural nets, and as such allows for model combination, at test time only a single network needs to be tested. By avoiding training all nodes on all training data, dropout decreases overfitting. The method also significantly improves training speed. This makes the model combination practical, even for deep neural networks. The technique seems to reduce node interactions, leading them to learn more robust features that better generalize to new data.

Convolutional neural network -- Part 2

Number of parameters Another simple way to prevent overfitting is to limit the number of parameters, typically by limiting the number of hidden units in each layer or limiting network depth. For convolutional networks, the filter size also affects the number of parameters. Limiting the number of parameters restricts the predictive power of the network directly, reducing the complexity of the function that it can perform on the data, and thus limits the amount of overfitting. This is equivalent to a "zero norm".

Convolutional neural network -- Part 3

Max norm constraints Another form of regularization is to enforce an absolute upper bound on the magnitude of the weight vector for every neuron and use projected gradient descent to enforce the constraint. In practice, this corresponds to performing the parameter update as normal, and then enforcing the constraint by clamping the weight vector $w \rightarrow \|\vec{w}\|_2 < c$ of every neuron to satisfy $\|\vec{w}\|_2 < c$. Typical values of c are order of 3–4. Some papers report improvements when using this form of regularization.

Convolutional neural network -- Part 4

For example, in CIFAR-10, images are only of size $32 \times 32 \times 3$ (32 wide, 32 high, 3 color channels), so a single fully connected neuron in the first hidden layer of a regular neural network would have $32 \times 32 \times 3 = 3,072$ weights. A 200×200 image, however, would lead to neurons that have $200 \times 200 \times 3 = 120,000$ weights. Also, such network architecture does not take into account the spatial structure of data, treating input pixels which are far apart in the same way as pixels that are close together. This ignores locality of reference in data with a grid-topology (such as images), both computationally and semantically. Thus, full connectivity of neurons is wasteful for purposes such as image recognition that are dominated by spatially local input patterns. Convolutional neural networks are variants of multilayer perceptrons, designed to emulate the behavior of a visual cortex. These models mitigate the challenges posed by the MLP architecture by exploiting the strong spatially local correlation present in natural images. As opposed to MLPs, CNNs have the following distinguishing features:

Convolutional neural network -- Part 5

Receptive field In neural networks, each neuron receives input from some number of locations in the previous layer. In a convolutional layer, each neuron receives input from only a restricted area of the previous layer called the neuron's receptive field. Typically the area is a square (e.g. 5 by 5 neurons). Whereas, in a fully connected layer, the receptive field is the entire previous layer. Thus, in each convolutional layer, each neuron takes input from a larger area in the input than previous layers. This is due to applying the convolution over and over, which takes the value of a pixel into account, as well as its surrounding pixels. When using dilated layers, the number of pixels in the receptive field remains constant, but the field is more sparsely populated as its dimensions grow when combining the effect of several layers. To manipulate the receptive field size as desired, there are some alternatives to the standard convolutional layer. For example, atrous or dilated convolution

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expands the receptive field size without increasing the number of parameters by interleaving visible and blind regions. Moreover, a single dilated convolutional layer can comprise filters with multiple dilation ratios, thus having a variable receptive field size.

Convolutional neural network -- Part 6

The spatial size of the output volume is a function of the input volume size W , the kernel field size K of the convolutional layer neurons, the stride S , and the amount of zero padding P on the border. The number of neurons that "fit" in a given volume is then:

Convolutional neural network -- Part 7

Fukushima's analog threshold elements in a vision model In 1969, Kunihiko Fukushima introduced a multilayer visual feature detection network, inspired by the above-mentioned work of Hubel and Wiesel, in which "All the elements in one layer have the same set of interconnecting coefficients; the arrangement of the elements and their interconnections are all homogeneous over a given layer." This is the essential core of a convolutional network, but the weights were not trained. In the same paper, Fukushima also introduced the ReLU (rectified linear unit) activation function.

Convolutional neural network -- Part 8

Drug discovery CNNs have been used in drug discovery. Predicting the interaction between molecules and biological proteins can identify potential treatments. In 2015, Atomwise introduced AtomNet, the first deep learning neural network for structure-based drug design. The system trains directly on 3-dimensional representations of chemical interactions. Similar to how image recognition networks learn to compose smaller, spatially proximate features into larger, complex structures, AtomNet discovers chemical features, such as aromaticity, sp³ carbons, and hydrogen bonding. Subsequently, AtomNet was used to predict novel candidate biomolecules for multiple disease targets, most notably treatments for the Ebola virus and multiple sclerosis.